



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
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Roll Number	2520090139
Branch	CSE-IT
Course Outcome	COURSE OUTCOME - 3

Case Study Topic: Minimum Spanning Tree (MST) for IoT Gateway Network Design using Kruskal and Prim Algorithms.

Work Done Summary

Implemented Minimum Spanning Tree (MST) algorithms to connect all IoT gateway rooms with minimum cable cost while avoiding cycles. Applied Kruskal's/Prim's algorithm to select the optimal network links and determine the minimum total installation cost.

Implementation Details

1. Represented the IoT gateway network as a weighted graph.
2. Defined gateway rooms: **A, B, C, D, E, F, G**.
3. Added all cable routes with their respective costs.
4. Sorted edges in ascending order of weight (Kruskal's approach).
5. Selected the minimum-cost edges while avoiding cycles.
6. Continued until all 7 gateway rooms were connected.
7. Constructed the Minimum Spanning Tree (MST).
8. Calculated the total cable installation cost of the MST.
9. Verified that every gateway room was connected with no redundant links.

Output

Minimum Spanning Tree:

E - F : 1

B - C : 2

F - G : 2

C - D : 3

A - C : 4

D - F : 4

Total MST Cost = 16

Result

1. Successfully constructed the Minimum Spanning Tree for the IoT gateway network.
2. Connected all seven gateway rooms without forming any cycles.
3. Selected the minimum-cost cable routes using Kruskal's/Prim's algorithm.
4. Reduced overall installation cost while maintaining full network connectivity.
5. Obtained an MST consisting of **6 edges** with a total cost of **17**.
6. Demonstrated the effectiveness of MST algorithms in network design and optimization.

Time Complexity

Operation	Time Complexity
Sorting Edges	$O(E \log E)$
Union-Find Operations	$O(E \alpha(V))$
Overall Complexity	$O(E \log E)$

Operation	Time Complexity
Building MST	$O(E \log V)$
Extract Minimum Edge	$O(\log V)$
Overall Complexity	$O(E \log V)$

Where:

- **V** = Number of vertices (gateway rooms)
- **E** = Number of edges (cable routes)

Conclusion

The Minimum Spanning Tree successfully connected all IoT gateway rooms with the minimum possible cable installation cost while avoiding redundant cycles. By selecting only the essential links, the MST reduced network deployment expenses and ensured efficient connectivity among all gateways. The final network achieved full coverage with a total cost of **17**, demonstrating that Kruskal's and Prim's algorithms are effective solutions for designing cost-efficient communication networks in IoT environments.

GitHub Link: <https://github.com/shashankreddy-21/DSA-Project>

Student Signature	Faculty Signature